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Predicting detailed body sizes by feature parameters

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Abstract

Purpose – Detailed body sizes are prerequisite for made to measure or customized manufacture. Nowadays, detailed body sizes can be precisely obtained by using 3D scanners, however, the high prices of the scanners block the population for such approaches. The purpose of this paper is to provide an economical and accurate data-driven method which can predict detailed body sizes with a small number of feature sizes.

Design/methodology/approach – First, the representative body sizes are extracted from dozens of detail body sizes by using factor analysis and garment knowledge. Among the representative body sizes, those that are easy to be measured are selected as the feature parameters (FPs). Second, by mining the database of the body sizes, mapping from the FPs to the detailed body sizes is expressed by a combination of radial basis function and multiply linear regression. Thus, for an individual human body, his/her detailed body sizes can be predicted by a small number of FPs.

Findings – First, FPs which are easily measured and represent the main shape information of a human body are extracted. Second, detailed body sizes can be functionally predicted by the FPs.

Originality/value – Traditionally, measuring dozens of body sizes for each human body is tiresome and the accuracy of the sizes depends on the experience of the gaugers. In this paper, a small number of body sizes are selected as the FPs which are easy to be measured and can functionally express the other body sizes. Thus, by only measuring the FPs, the detailed body sizes can be intelligently and automatically predicted. This approach is meaningful to improve the intelligence and accuracy of the measurement, so that even an inexperienced gauger is competent to obtain accurate detailed body sizes.

Keywords Body sizes prediction, Data driven, Feature parameters

Paper type Research paper

1. Introduction

Fitness is a key factor affecting people's dressing. Made to measure (MTM) or customization manufacture (CM) is a good way to improve the clothing fitness, in which the patterns are designed according to the individual figures of the customers. Accurate and detailed body sizes are required in MTM or CM.

Traditionally, the body sizes are measured manually by gaugers with tapes. The main disadvantage of such methods is that the accuracy of the measurement subjectively depends on the experience of the gaugers. Usually, it takes years for a

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gauger to become an expert. Additionally, it is also annoying to manually measure dozens of body sizes for each customer. Nowadays, 3D scanners make it possible to get body size quickly and detailedly, however, their high prices are not acceptable for most enterprises and consumers.

How to accurately and economically get the detailed body sizes is still an open problem in the clothing industry. For addressing this issue, a data-driven-based body size measurement is proposed in this paper. Our method is based on the facts that for each individual human body, correlations exist among his/her different body sizes, and the correlations obey certain statistical rules. Thus, if we can extract the key or the feature body sizes, and use these feature body size to mathematically express the other sizes, we can obtain the following benefits.

On the one hand, the workload of the manual measurement can be greatly reduced; on the other hand, we use the data of detailed body sizes measured by the 3D scanners to train the correlation functions, therefore the accuracy of the 3D scanners can be shared. For example, the feature size such as waist girth is more possible to get an accurate measurement than crotch length for an inexperienced gauger, thus if we use the feature sizes to functionally predict the other detailed body sizes, the error of measurement could be reduced. Such technology can benefit the MTM industry.

Figure 1 shows the framework of this method. Based on the anthropometric database with its data obtained by 3D body scanning, we first analyze the correlations between variables in the database, and extract the feature parameter (FP) by using factor analysis (FA). Second, we set the mapping from the FPs to the other body sizes by a combined prediction model that is formed by non-linear radial basis function (RBF) and multiply linear regression (MLR). The weights on RBF and MLR are assigned by applying induced order weighted average (IOWA) operator. With the mapping function, the detailed body sizes for an individual human body can be predicted by inputting his/her FPs.

The rest of this paper is organized as follows. Section 2 introduces the related works; Section 3 extracts the FPs from the detailed body sizes; Section 4 builds the relationship between the FPs and the detailed body sizes based on combined predication method; Section 5 gives the experimental results and analysis; Section 6 concludes the whole work and points out the future research.

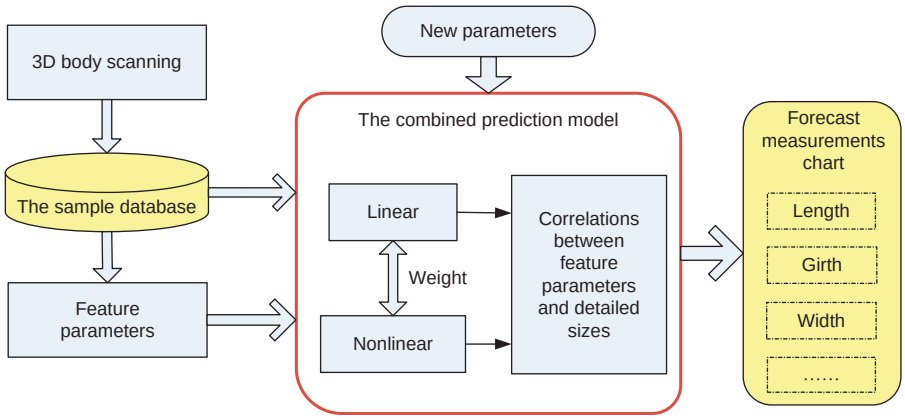

Figure 1. Framework of combined forecast method

2. Related work

Body sizes measurement is the fundamental for clothing design and manufacture, especially for MTM. Oliver and Shim (1993) profiled MTM clothing for male users, and presented the anthropometric data collection that was the premise of the MTM. Mpampa *et al.* (2010) gave a new size system that can classify people effectively according to the defined several body character measurements. With the development of information and network technology, the MTM procedure has been improved. Cordier *et al.* (2003) presented a system of e-MTM, including body and clothing creation, clothed human body simulation and online sizing. Garment visualization is the last step in MTM procedure, 3D clothing simulation is a difficult point for the step. Chen (2009) gave an approach which can enhance clothing visualization to modeling 3D woven-shaped fabric. Recently, Chen *et al.* (2011) provided an overview of the clothing modeling, simulation and visualization methods. These works mainly focussed on the MTM process technology. The measurement data are collected from 3D body scanner.

Some works analyzed the parameters of human body figure, most of which used linear method to classify people's figure. The FA and principal component analysis (PCA) are common ways to decrease the data dimensions. Hsu (2009) transferred the anthropometric variables to some factors through FA, clustered body types by K-means algorithm, and gave the measurements data distribution for each type. Zheng *et al.* (2007) presented a new bra size chart by PCA and K-means clustering. Chu *et al.* (2010) extracted the information from 3D body meshes using PCA, and modeled the correlation between the feature measurement parameters and the principal component information. Their method can rebuild the 3D body when inputting a new feature of data.

Nevertheless, the relationship among the anthropometric variables is not completely linear. Neural net has a strong ability of non-linear approximation and can be a good substitute for the linear. Lu and Wei-yan (2008) used back propagation (BP) net to model the relationship between the character parameters and the feature curve in order to build a 3D parameter body. Hui and Ng (2009) predicted the seam performance of commercial woven fabric using neural net. BP net is a common tool for prediction and fitting. Although it could obtain good results, the convergence speed is slow. RBF net improves this defect well. Some researcher (Nurwaha and Wang, 2012; Lin *et al.*, 2009) take RBF net used for forecast issues. Nadia *et al.* (2011) deformed the 3D body model using RBF based on selected feature measurements.

The type of correlations between anthropometric variables can be divided into three classes, comprising strong linear, weak linear and non-linear correlations. Using only one type of model to predict the figure detailed data would lose information for some variables inevitably. The concept of combining predicts started with the seminal work 50 years ago (Bates and Granger, 1969). Combined model and multiply level prediction can enhance the accuracy for predicting effectively (Costantini and Pappalardo, 2010). Chen *et al.* (2011) presented a linear-non-linear model to predict the requirement of tourism demands.

3. FPs extraction

FP is extracted from the anthropometric variables in the database. They are the input section in the method, and impact the accuracy of the predicating sizes directly. It is a pivotal step that how the FPs are selected. Choosing too many parameters would result

in serious collinearity between the FPs, which will cause collinearity interference during prediction. Choosing too few parameters would lead to prediction deviation because the FPs have not enough information. We transfer the anthropometric variables as several factors with FA.

Suppose that the sample $X = (x_{ij})_{m \times p}$ has m variables in p dimensions, Y is the normalization of X, U and $\lambda = diag\ (\lambda_1, \lambda_2, \ldots, \lambda_p)$ are the eigenvectors and eigenvalues of $C = Y \times Y^T$, i.e. $C = U\lambda U^T$.

Set $Q \times Q^T = \lambda$, the factor matrix can be indicated as $Q = U^T Y$, where $Q_j = U^T Y_j$, $j = 1, 2, \ldots p$, is factor value for the jth measurement.

FA describes data with public factor and special factor. The public factor is defined by the contribution of variables variance. According to contribution's ranking table, we take the k ($k < p$) factors that the accumulated eigenvalue is comparatively great enough.

$\sum_{i=1}^{k} \lambda_i / \sum_{i=1}^{p} \lambda_i \geqslant R$, where R is a value in range [0.7, 0.9].

Thus, the matrix Y can be expressed by:

$$Y = UQ + \varepsilon$$

where $UQ = U_{p \times k}\ Q_{k \times m}$ is main or public factors, $\varepsilon = U_{p \times (p-k)}\ Q_{(p-k) \times m}$ is special factors. The component coefficients matrix is $A = (a_{ij})_{k \times k}, a_{ij} = U_{ij}\sqrt{\lambda_{ij}}$. The factors model are following as: $Y = AQ' + \varepsilon$, $E(\varepsilon) = 0$, where $Q' = Q/\sqrt{\lambda}$.

We can know the relationship which the anthropometric variables mean to the factors through component coefficients matrix A.

Suitability test for the sample had been done before FA. The Kaiser-Meyer-Olkin's measure of sampling adequacy is 0.88, and Bartlett's test is valid ($p < 0.001$). According to initial eigenvalues cumulative shown as Table I, we select six factors as variables index ($R = 0.8$). These independent factors can express the whole information mostly.

With the rotated component matrix for each factor, we get the variables coefficients for corresponding factors. As the Table II shows, we name the six factors based on anthropometric and garment knowledge.

The component coefficient represents the degree of variables concentrated within the factors. The higher coefficient means that the variable is more important to the factor, and vice versa. We take the high score coefficient variables as the character parameters candidates. In consideration of different factor's eigenvalues variance contribution as Table II, two candidates are selected in factor 1 and factor 2, one

No.	λ_i	$\lambda_i/\sum\lambda_j$ (%)	Accumulation (%)
1	19.360	40.334	40.334
2	11.043	23.007	63.340
3	2.732	5.691	69.031
4	2.421	5.044	74.076
5	1.883	3.924	77.999
6	1.377	2.869	80.868
7	1.187	2.473	83.342
8	1.136	2.368	85.709
9	0.827	1.723	87.432

Table I. Eigenvalues explained of anthropometric variables

Table II.
Factor name and the high rotated component coefficients measurements

Order	Factor name	Measurements with high rotated component coefficients
1	Height	Belly height (0.9857), hip height (0.9824), inseam (0.9788), waist height (0.9735), neck height (0.949)
2	Girth	Breast girth (0.9203), belly girth (0.8867), upper arm girth (0.8710), waist girth (0.8743), hip girth (0.6859)
3	Small joint girth	Wrist girth (0.7447), knee girth (0.7429), ankle girth (0.7218), mid-neck girth (0.6480)
4	Waist-hip	Waist to high hip back (0.9523), distance waist to hip (0.8936), distance neck to hip (0.5742)
5	Upper torso	Back length (0.8083), side upper torso length (0.7947)
6	Shoulder	Shoulder width (0.8292), cross shoulder over neck (0.6319)

candidate is selected in factors 3-6. The FPs candidates are filtered by the consideration that the parameters should fit for anthropometric rules and meet the size system demands. The specific principles are as follows:

(1) FP can be measured by manual method conveniently with low errors. Body size such as belly is difficult to define exactly for people. In factor 1 and factor 2, the belly height and girth are instead of inseam and waist girth. Similarly, the distance from waist to high hip back that measures the vertical distance from waist line to crista iliaca has a high coefficient, thus distance from waist to hip is a substitute for this parameter because few people know crista iliaca location.

(2) FP should be representative for the body shape variation regulation. Waist height and hip height get a high score coefficient in height factors. However, the neck height reflects human body length more obviously. So we take neck height and inseam length as height FP.

(3) FP should be compatible for existing size system and clothes making. In factors 3 and 5, side torso length and back length score is close, wrist girth and knee girth score is close too. Considering the size system, we selected back length and wrist girth as parameters.

As Table III shows, the FPs are: neck height, inseam length, breast girth, waist girth, distance waist to hip, wrist length, back length and shoulder width.

Table III.
FPs for each factor

Factors	Height		Girth		Small joint girth	Waist hip	Upper torso	Shoulder
FPs								
	Neck height	Inseam	Breast girth	Waist girth	Wrist girth	Distance waist to hip	Back length	Shoulder width

4. Relationship between FPs and other body sizes

4.1 Sizes prediction with linear and non-linear model

The prediction for detailed body size is a mapping: $\varphi(x):\mathbb{Z}^p \rightarrow \mathbb{R}^q$ which minimizes $\|\hat{y} = \varphi(x) - y\|$ with y being the real data. For different types of anthropometry variables in database, it is inevitable to make a large error with single method. We propose a combined method as the following equation:

$$Y = [y_1, \cdots, y_i, \cdots, y_n]^T, \quad y_i = \sum_{j=1}^{m} W_j \varphi_j(b_i) \quad i = 1, 2, \cdots n; j = 1, 2, \cdots m \tag{1}$$

where $\varphi_j(b_i)$ is the jth kind method ith variable data according to input FPs. W_j is the weight of jth kind of methods. In our approach, the predicating model combines MLR and RBF, so as to fit the linear and non-linear correlation between the FPs and detailed body sizes.

Figure 2 shows the frame of RBF net that constituted input layer, hidden layer and output layer. The input layer is the inputting FP. The hidden layer constructs a hidden space that mapping the inputting low dimensional parameters to high dimensional node data based on function Φ. The output layer output the result r through linear-weighted summation method. The function Φ plays a critical effect during the inputting FP, called the basis function.

In this paper, suppose the number of training samples is n. The input layer variable is the FP vector b with k dimension. The output layer is the prediction of measurement $r(b)$. The hidden layer gives b a non-linear transformation-based ϕ function. The process is provided in detail as follows:

$$r(b) = \sum_{i=1}^{s} w_i \Phi(\|b - C_i\|), \quad \Phi(\|b - C_i\|) = \exp\left(-\frac{\|b - C_i\|^2}{2\sigma_i^2}\right), \quad i = 1, 2, \cdots, s \tag{2}$$

where w_i is the output weight in the linear weighted summation, s is the number of the hidden layer nodes. $\Phi(\cdot)$ is the basis function, $\|\cdot\|$ is the Euclidean distance. C_i is the center of the hidden node, and its dimension equals to the input vector. σ_i is the width of the RBF unit. The basic function uses the Gaussian activation function.

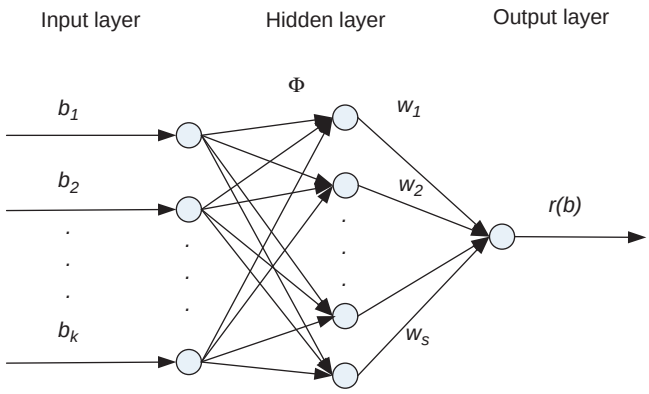

Figure 2.
Frame of the RBF net

The MLR method can be expressed as:

$$r(b) = \beta_0 + \beta_1 b_1 + \cdots + \beta_k b_k + \varepsilon \tag{3}$$

where ε is the random error, β is the coefficient, and we suppose $E(\varepsilon) = 0$ generally. If the number of training sample is greater than k, β can be solved through least square method theoretically.

4.2 Weights of the combined prediction model based on IOWA

The prediction precision of combined model is decided by component methods precisions and distributed weights. Weights can improve the prediction accuracy with complementary advantages. Yager and Filev (1999) introduced the IOWA operator to provide a method for aggregating argument values. IOWA operator assigns the weights of the method according to prediction accuracy at different time. In this paper, the process of the operator is given as:

Suppose p_{jt} is the prediction accuracy of jth kind method at sample t, $P_{jt} \in [0,1]$. Then, p_{jt} is the induced order of prediction value r_{jt}. We can get an m number array to describe the jth kind method prediction at sample t as follows:

$$IOWA(\langle p_{1t}, r_{1t}\rangle, \langle p_{2t}, r_{2t}\rangle, \cdots \langle p_{mt}, r_{mt}\rangle) = \sum_{j=1}^{m} W_j r_{p-index(jt)} \tag{4}$$

where $p-index(jt)$ is the j subscript of the larger predicting accuracy at sample t, applies the ordering of the argument values. $W_j = (w_1, w_2, \cdots w_m)^T$ is the ordered weighted average coefficient vector of the jth prediction methods.

Set $e_{p-index(ji)} = r_i - r_{p-index(ji)}$, the sum of error square S for the total sample is given as:

$$S = \sum_{t=1}^{N} \left(r_t - \sum_{j=1}^{m} W_j r_{p-index(jt)} \right)^2 = \sum_{j=1}^{m} \sum_{l=1}^{m} W_j W_l \left(\sum_{t=1}^{N} e_{p-index(jt)} e_{p-index(lt)} \right) \tag{5}$$

Then we obtained the weights for methods through the following equation:

$$\min S(W_1, W_2, \cdots W_m) = \sum_{j=1}^{m} \sum_{l=1}^{m} W_j W_l \left(\sum_{t=1}^{N} e_{p-index(jt)} e_{p-index(lt)} \right)$$
$$s.t. \quad \begin{cases} \sum_{j=1}^{m} W_j = 1 \\ W_j \geqslant 0, j = 1, 2, \cdots m \end{cases} \tag{6}$$

5. Experimental results and analysis

We build a sample set included 400 female college students aged 18-24. The 3D body scanner measured over 60 dimensions data for each student. The whole sample set was divided into two parts. One part is for training and the other part is for testing. To avoid the interference caused by different body type, the two parts were picked up through systematic method from the sample sorted by body height and breast girth.

For evaluating the prediction accuracy, the following distance metrics are employed: root mean squared error (RMSE) and mean absolute percentage error (MAPE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{7}$$

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \times 100\% \tag{8}$$

where y_t and $\hat{y}_t$ are the actual and predicted value, respectively. While both MAE and RMSE are simply measurements of discrepancies between the predicted values and the actual observations, MAPE measures scaled discrepancies at each t.

To evaluate our prediction objectively, we compared MLR, RBF, PCA reverse and combined prediction model to get prediction results. The PCA reverse method refers to Chu *et al.* (2010).

First, we had a prediction test for methods in different samples. Body height, breast thickness and crotch length were picked as the test objects. As Table IV shows, the Person coefficients between FP and the test objects demonstrates body height has a high linear relation with the parameters, and the other two have a small correlation.

Table V shows the test results. PCA method gives an error value of RMSE higher than the others. The reason is that PCA abandons little components information when extract the principal components. The information between the non-linear measurements and the parameters is lost during building correlation for the principal components and the parameters with least square method. The result caused by two reasons above in magnified value of RMSE.

	Neck height	Shoulder width	Breast girth	Waist girth	Back length	Distance waist to hip	Inseam	Wrist girth
Body height	0.990	0.100	0.118	0.005	0.644	0.466	0.926	−0.097
Breast thickness	0.210	0.1596	0.912	0.714	0.138	0.060	0.175	0.283
Crotch length	0.482	0.140	0.409	0.484	0.161	0.573	0.294	0.199

Table IV. The Person correlations coefficients between objects and FP

Training sample	Test sample	Body height RMSE			Breast thickness RMSE			Crotch length RMSE		
		MLR	PCA	RBF	MLR	PCA	RBF	MLR	PCA	RBF
100	50	1.3532	10.2059	2.5907	1.8488	131.0586	1.2270	5.2086	186.0283	3.0463
150	50	1.3374	9.8239	2.5831	1.8393	151.3736	1.0679	5.0089	196.4651	2.8155
200	50	1.3344	9.9321	2.3286	1.7946	148.1657	1.1406	5.0277	174.3865	2.7050
250	50	1.3344	10.7365	2.3870	1.8004	137.8376	1.0568	4.9464	177.3671	2.6462
300	50	1.3435	9.4242	2.1178	1.7965	149.3714	0.9792	4.9321	169.3642	2.5304
350	50	1.3458	10.1982	2.0842	1.8088	154.2732	0.9518	4.9277	179.0275	2.4270

Table V. RMSE for different methods under increased sample condition

In other methods, the MLR is better than RBF in body height with high linear relation, and the RBF is better than MLR in breast thickness and crotch length. As the training samples growing, the accuracy of RBF was improved accordingly, and the MLR and PCA maintained the same level.

We distributed the sample as a 320-training sample and a 80-test sample. We trained the sample and got the test predicting results for detailed body sizes with MLR, RBF and the combined method separately.

As the RMSE and the MAPE value graphs shown in Figures 3 and 4, the RBF is better than MLR in girth, thickness and width measurements. This predicting precision is obviously higher for non-linear method such as in shoulder angle and crotch length measurements. MLR is better than RBF in some height measurements such as distance neck to knee and side seam height.

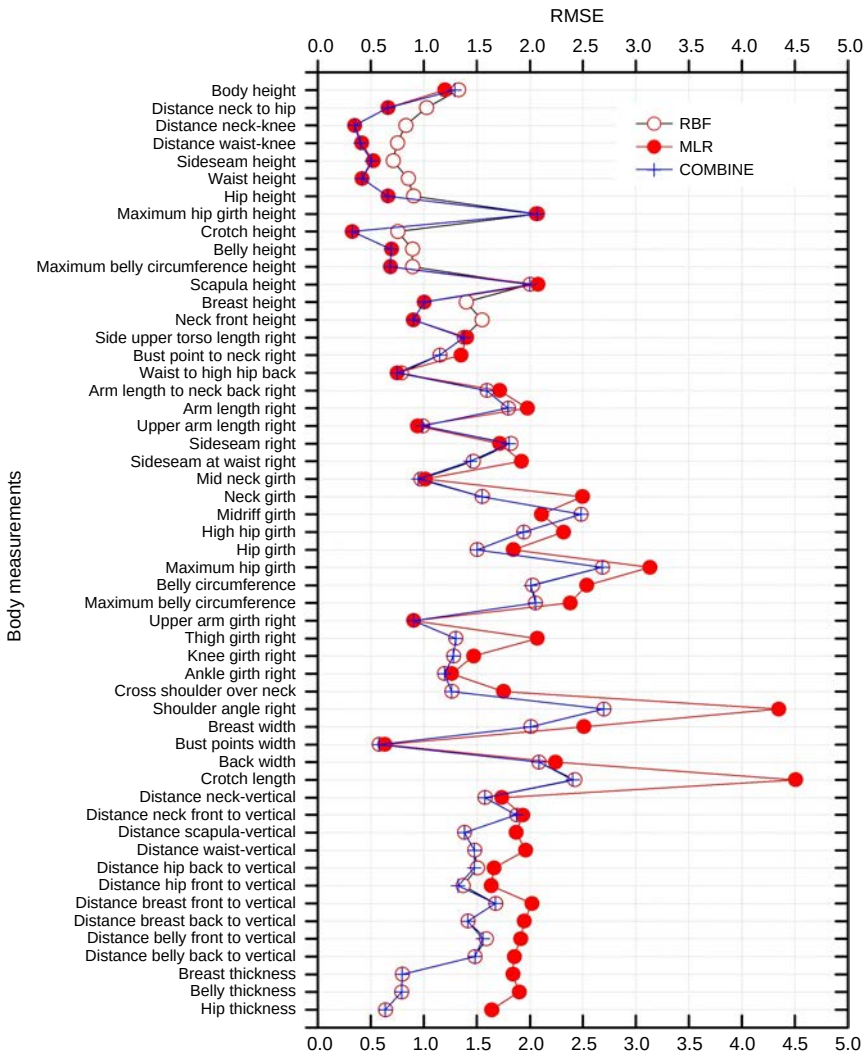

Figure 3.
RMSE values for detailed body sizes test sample

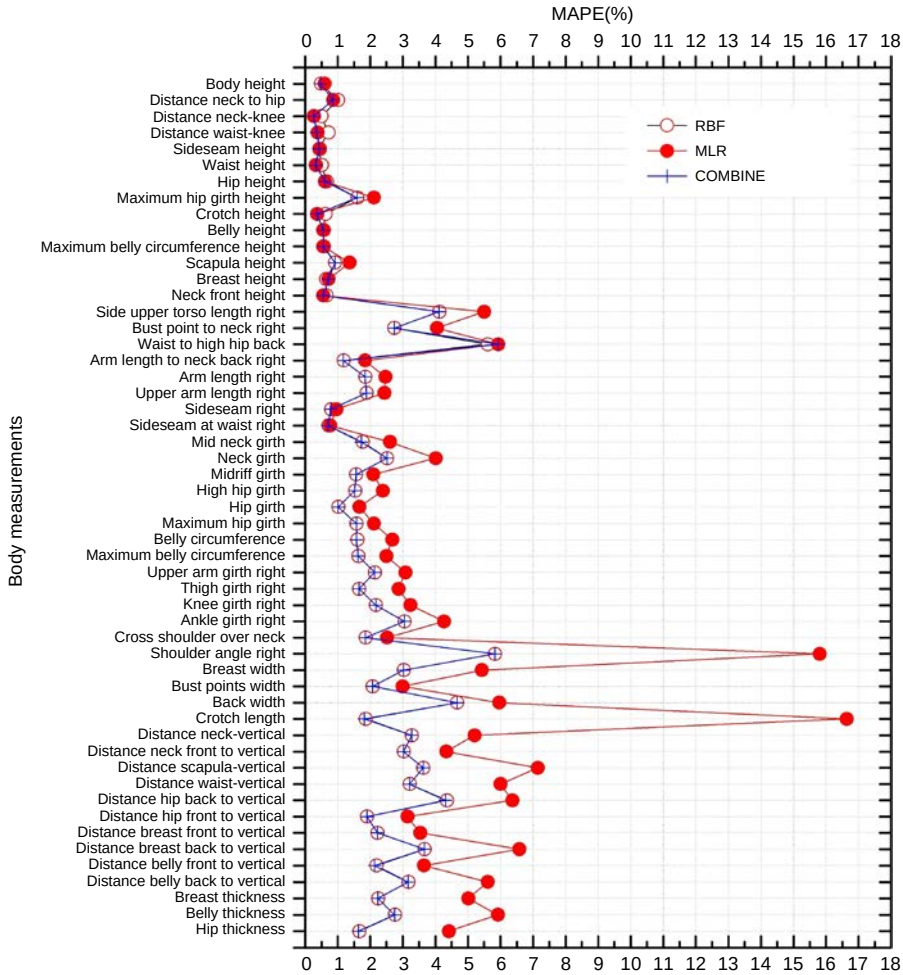

Figure 4.
MAPE values for detailed body sizes test sample

The combined predicting value is the best of the all methods no matter in RMSE or MAPE. And the IOWA weights in the combined method are shown in Table VI. We can find that the weights give the all distribution to the better methods in many measurements. The combined predicting results are the better one between RBF and MLR. This is the reason that the prediction value of two methods is higher or lower than the actual value for the same. This indicates that they have poor complementarity. If the weight gave the distribution, the final results would be more deviated from the actual value. The best way is to choose the better one as the combined result.

6. Conclusions

This paper present a method for predicting detailed body sizes with a small number of FPs. The FPs are selected based on FA analysis and the knowledge of anthropometric and garment. The predicting model is a combination of linear MLR and non-linear RBF methods with their weights computed by IOWA operator. The experiments indicate

Table VI. IOWA weights for RBF and MLR methods in combined model

Measurements	IOWA weight RBF weight	IOWA weight MLR weight	Measurements	IOWA weight RBF weight	IOWA weight MLR weight
Body height	0.9762	0.0238	Maximum hip girth	1	0
Distance neck to hip	0	1	Belly circumference	0.9779	0.0221
Distance neck-knee	1	0	Maximum belly circumference	1	0
Distance waist-knee	0.0285	0.9715	Upper arm girth right	1	0
Side seam height	0.0306	0.9694	Thigh girth right	1	0
Waist height	0	1	Knee girth right	1	0
Hip height	0	1	Ankle girth right	1	0
Maximum hip girth height	1	0	Cross shoulder over neck	1	0
Crotch height	0	1	Shoulder angle right	1	0
Belly height	0	1	Breast width	1	0
Maximum belly circumference height	0	1	Bust points width	1	0
Scapula height	1	0	Back width	1	0
Breast height	0	1	Crotch length	0.9639	0.0361
Neck front height	0	1	Distance neck-vertical	1	0
Side upper torso length right	1	0	Distance neck front to vertical	1	0
Bust point to neck right	1	0	Distance scapula-vertical	1	0
Waist to high hip back	0	1	Distance waist-vertical	1	0
Arm length to neck back right	1	0	Distance hip back to vertical	0.9753	0.0247
Arm length right	1	0	Distance hip front to vertical	0.9439	0.0561
Upper arm length right	0	1	Distance breast front to vertical	1	0
Side seam right	0.9811	0.0189	Distance breast back to vertical	1	0
Side seam at waist right	0.0289	0.9711	Distance belly front to vertical	0.9691	0.0309
Mid-neck girth	0.9715	0.0285	Distance belly back to vertical	1	0
Neck girth	1	0	Breast thickness	1	0
Midriff girth	1	0	Belly thickness	1	0
High hip girth	1	0	Hip thickness	0.9633	0.0367
Hip girth	1	0			

that the combined method can improve the accuracy of the predictions. The predicting value can be used for MTM or CM directly.

The results are acquired based on the sample data, requiring the figure in test database should be within the shape space of the sample database. Currently, we only have a limited female college students' data for variety. In the future, we will focus on how to improve the prediction accuracy and the adaptability of the model for different types of people.

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